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Adaptive Performance-Weighted Signal Ensemble: A Comparative Study of Portfolio Performance

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Abstract

The thesis investigates whether an adaptive ensemble of trading signals can outperform a traditional equal-weighted portfolio in terms of return and risk-adjusted performance. Financial markets are characterized by uncertainty, non-stationarity, and varying market regimes. This makes it difficult for any single strategy to deliver strong performance consistently. To address this challenge, the study proposes an Adaptive Performance-Weighted (APW) framework that dynamically assigns weights to trading signals based on their recent predictive accuracy.

The proposed model is evaluated using a backtesting framework applied to a diversified universe of 65 exchange-traded funds (ETFs) across multiple asset classes, including equities, fixed income commodities, and currencies. The analysis covers the period from 2017 to 2025 and compares the APW model with a baseline Equal-Weighted (EW) portfolio. Performance is assessed using both return-based and risk-adjusted metrics, including annual returns, volatility, Sharpe ratio, Sortino ratio and maximum drawdown.

The empirical results indicate that the APW model can improve return generation in several periods across trending market environments. The adaptive weighting mechanism allows the model to allocate more capital to signals that have recently performed well, resulting in higher returns in certain years than the equal-weighted benchmark. However, the results also show that this improvement is not consistent across all market conditions. During periods of market instability, both portfolios experience similar levels of drawdown and volatility.

In terms of risk-adjusted performance, both strategies show limitations, as reflected in the presence of negative Sharpe and Sortino ratios in several years. This suggests that although the APW model can enhance return potential, it does not eliminate the trade-off between return and risk. The analysis further reveals that the adaptive model introduces greater variability in returns.

A robustness analysis shows that model performance is sensitive to parameter choices of lookback window, prediction horizon, and weighting scheme. No single configuration is found to be universally optimal. This highlights the importance

of balancing adaptability and robustness in portfolio design.

Keywords

Adaptive weighting, ensemble methods, momentum, portfolio optimization, risk-adjusted return, Sharpe ratio, Sortino ratio, Trend-following, Signal aggregation, volatility

Abstract

Detta examensarbete undersöker om en adaptiv ensemble av handelssignaler kan överträffa en traditionell likaviktad portfölj med avseende på avkastning och riskjusterad prestation. Finansiella marknader präglas av osäkerhet, icke-stationaritet och varierande marknadsregimer, vilket gör det svårt för en enskild strategi att konsekvent leverera goda resultat. För att hantera detta problem föreslår studien en adaptiv modell, Adaptive Performance-Weighted (APW), som dynamiskt viktar handelssignaler baserat på deras senaste prediktiva förmåga.

Modellen utvärderas genom en backtesting-metod på en diversifierad portfölj bestående av 65 börshandlade fonder (ETF:er) inom flera tillgångsslag, inklusive aktier, räntor, råvaror och valutor. Analysen omfattar perioden 2017–2025 och jämför den adaptiva modellen med en baslinjemodell med lika vikter (Equal-Weighted, EW). Prestationen utvärderas med både avkastningsbaserade och riskjusterade mått, såsom årlig avkastning, volatilitet, Sharpekvot, Sortinokvot och maximal nedgång (drawdown).

Resultaten visar att APW-modellen kan förbättra avkastningen under vissa perioder, särskilt i marknadsmiljöer med tydliga trender. Genom att ge större vikt åt signaler som nyligen har presterat väl kan modellen bättre utnyttja gynnsamma marknadsförhållanden. Samtidigt visar resultaten att denna förbättring inte är konsekvent över alla perioder. Under turbulenta marknadsförhållanden uppvisar båda portföljerna liknande nivåer av volatilitet och nedgångar.

Ur ett riskjusterat perspektiv uppvisar båda strategierna begränsningar, vilket framgår av negativa Sharpe- och Sortinokvoter under flera år. Detta indikerar att även om den adaptiva modellen kan förbättra avkastningspotentialen, eliminerar den inte den grundläggande avvägningen mellan avkastning och risk. Analysen visar också att den adaptiva modellen leder till större variation i avkastningen, vilket innebär en avvägning mellan anpassningsförmåga och stabilitet. En robusthetsanalys visar att modellens prestation är känslig för val av parametrar, särskilt avseende lookback-fönster, prediktionshorisont och viktning. Ingen enskild parameterkonfiguration visar sig vara optimalt i alla marknadsmiljöer,

vilket understryker vikten av att balansera flexibilitet och stabilitet.

Sammanfattningsvis visar studien att adaptiv viktning kan ge relativa förbättringar jämfört med likaviktning under vissa marknadsförhållanden, men att den inte konsekvent överträffar i alla situationer. Resultaten tyder på att adaptiva ensemblemetoder är en lovande ansats inom portföljkonstruktion, men att deras effektivitet beror på marknadsregim och signalernas stabilitet

Nyckelord

Adaptiv viktning, ensemblemetoder, momentum, portföljoptimering, riskjusterad avkastning, Sharpekvot, Sortinokvot, Trendföljandestrategi, Signalaggregering, volatilitet

Terminology

Adaptive Performance-Weighted (APW): A portfolio construction method where trading signals are dynamically weighted based on their recent predictive performance.

Equal-Weighted (EW): A baseline portfolio strategy where all signals are assigned equal weights.

Exchange-Traded Fund (ETF): An investment fund that is traded on stock exchanges, they are typically tracking an index, commodity, or basket of assets.

Idiosyncratic Risk: Risk that is specific to an individual asset or signal and not explained by overall market movements.

Momentum Strategy: An investment approach that exploits trends by assuming that assets with strong past performance will continue to perform well in the near future.

Trend-Following: A strategy that tries to detect gains by identifying and following sustained movements in asset prices.

Signal Aggregation: The process of combining multiple trading signals into a single decision or portfolio allocation

Volatility: A statistical measure of the dispersion of returns, commonly used as an indicator of risk.

Sharpe Ratio: A measure of risk-adjusted return defined as the excess return per unit of total volatility.

Sortino Ratio: A risk-adjusted performance measure that considers only downside volatility.

Drawdown: The decline in portfolio value from a peak to a subsequent trough.

Lookback Window: The historical period used to evaluate the past performance of signals.

Prediction Horizon: The forward time period over which returns are predicted.

Backtesting: The process in which you evaluate a strategy using historical data to simulate its past performance.

Portfolio Optimization: The process of selecting the best allocation of assets or signals to maximize the return for a given level of risk.

Multi-Asset Portfolio: A portfolio that has different types of assets, such as equities, bonds, commodities, and currencies.

Liquidity: The ease with which an asset can be bought or sold in the market without significantly affecting its price.

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Use of Digital Tools

In the final stage of the thesis-writing process, Grammarly was used as a digital language review tool to check and improve grammar, spelling, punctuation, and linguistic clarity. The tool was used only for proofreading and language correction. It was not used to generate new content, alter the analysis, or influence the results and conclusions of the thesis. All suggested changes were reviewed and approved by the authors.

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1 Introduction

Financial markets are inherently complex and dynamic. They are characterized by a high degree of uncertainty. Asset prices are influenced by a wide range of factors such as macroeconomic conditions, investor sentiment, and other unexpected external shocks such as wars. As a result, predicting future returns is a challenging task, and no single investment strategy consistently outperforms others across different market regimes.

Traditional portfolio allocation methods rely on estimates of expected returns and risk. But these approaches are often highly sensitive to estimation error. They often lead to unstable portfolio weights and poor out-of-sample performance. At the same time, simple approaches such as equal weighting provide robustness. But they lack the ability to adapt to changing market conditions. This creates a challenge in portfolio construction about how to balance stability and flexibility under uncertainty.

Previous research suggests that combining multiple strategies can improve performance by capturing different aspects of the market. There remains a gap in understanding how strategies can be combined using simple and adaptive data-driven methods.

This thesis addresses this problem by proposing an Adaptive Performance-Weighted (APW) framework. This is a method where multiple trading strategies are dynamically weighted based on their recent performance. This approach is evaluated using a multi-asset dataset of exchange-traded funds (ETFs). This will be done over the period 2017-2025. The performance of the APW model is compared to a benchmark Equal-Weighted (EW) strategy using both returns and risk-adjusted metrics. The main objective of this study is to investigate whether adaptive signal weighting can improve risk-adjusted portfolio performance. This is compared to static allocation approaches.

The remainder of this thesis is structured as follows. Chapter 2 presents the theoretical background and related work on financial prediction, portfolio models, ensemble models, risk-adjusted performance, uncertainty, and model robustness. Chapter 3 introduces the theoretical foundations of the study, including

momentum and trend-following, diversification, ensemble method, and risk-adjusted performance measures. Chapter 4 describes the methodology, including data collection, signal construction, model parameters, weighting schemes, and comperes the performance of the backtesting framework. Chapter 5 presents the empirical results and compares the performance of the Adaptive performance-Weighted model with the Equal-Weighted benchmark using return-based and risk-adjusted measures. Chapter 6 discusses the findings, robustness analysis, reliability, limitations, directions for future research and final conclusions.

2 Background and Related Work

This chapter presents the theoretical background relevant to the study. It situates the proposed approach within existing research on financial prediction models, portfolio optimization, ensemble methods, and uncertainty in financial decision-making.

2.1 Financial Prediction and Portfolio Models

A foundational framework in portfolio theory is the mean-variance approach introduced by (Markowitz, 1952), where investors aim to balance expected return against risk. However, in practice, such models rely on estimates of errors. Empirical research shows that the errors in the estimation of expected returns can lead to unstable portfolio weights and poor out-of-sample performance. Even simplified approaches such as minimum-variance portfolios remain sensitive to estimation error and may produce unstable allocations over time.

To address these limitations, alternative approaches have been developed. Robust portfolio optimization methods aim to reduce sensitivity to model misspecification and improve stability (DeMiguel Nogales, 2009). For example, robust estimators can produce portfolio weights that are less sensitive to deviations from assumed distributions. Thereby improving out-of-sample robustness (DeMiguel Nogales, 2009). Similarly, Bayesian approaches incorporate prior information and allow models to adapt to changing market conditions. This is important in environments characterized by time-varying volatility (Bodnar et al., 2021).

Another important approach is rule-based trading strategies. Momentum and trend-following strategies are among the most widely studied and applied approaches. These strategies exploit the empirical observation that assets exhibiting positive (negative) past performance tend to continue performing well (poorly) over certain horizons. Evidence from long-term studies shows that such strategies generate persistent positive returns and low correlation with traditional asset classes. Furthermore, combining multiple signals or asset classes has been shown to improve risk-adjusted performance compared to relying on a single

strategy (Brooks, 2017).

2.2 Ensemble Methods in Finance

Ensemble methods are based on the idea that combining multiple models can lead to better performance than relying on a single model. This is relevant in financial markets where different strategies tend to perform well under different conditions.

In machine learning, ensemble methods such as bagging and boosting are widely used to improve predictive accuracy and robustness. However, in financial applications, the focus is often combining fundamentally different strategies rather than identical models trained on different data samples.

(Rodriguez Dominguez et al., 2025) propose a multi-hypothesis prediction framework in which each predictor corresponds to a different asset or hypothesis, and the ensemble combines these predictions into a portfolio decision. Their results show that diversity among predictors plays a key role in improving out-of-sample performance and enhancing risk diversification (Rodriguez Dominguez et al., 2025). Importantly, they establish a direct link between diversity in model performance and enhancing risk diversification in portfolio construction.

A natural benchmark for an ensemble approach is an equally weighted portfolio, where all strategies contribute equally regardless of past performance. While equal weighting is simple and robust, but it lacks alignment, which may limit its effectiveness in dynamic market environments.

2.3 Risk-Adjusted Performance

Evaluating trading strategies takes more than just comparing returns. A strategy with high returns may also involve high risk, making it less attractive from an investment perspective. Therefore, performance is typically assessed using risk-adjusted measures.

The Sharpe ratio is one of the most widely used metrics for this purpose. It measures the excess return of a strategy relative to its volatility. This provides an indication of how efficiently risk is being compensated. Many studies use

the Sharpe ratio to demonstrate the robustness and consistency of returns across different periods of time (Hedlund, 2024).

However, high in-sample Sharpe ratios do not translate into strong out-of-sample performance (Lassance et al., 2023). Parameter uncertainty and estimation error can lead to overly optimistic results during model calibration. Research shows that portfolio performance is highly sensitive to these uncertainties and that even well-performing strategies may fail to generalize (DeMiguel Nogales, 2009).

An alternative measure is the Sortino ratio, which focuses on the downside risk rather than total volatility. This can provide a more nuanced view of performance for strategies with asymmetric return distributions.

2.4 Uncertainty and Model Robustness

Uncertainty is a challenge in financial modeling. Models are typically estimated using historical data, but future market conditions may differ significantly from the past. This creates a gap between in-sample optimization and out-of-sample performance (Lassance et al., 2023).

Several studies highlight the importance of accounting for parameter uncertainty in portfolio decisions. Strategies optimized based on historical estimates may perform poorly when applied to new data due to randomness in both parameter estimation and realized returns. This suggests that relying on a single model or strategy may be risky (DeMiguel Nogales, 2009).

Robust approaches attempt to mitigate this issue by incorporating uncertainty directly into the modeling process (DeMiguel Nogales, 2009). Bayesian methods allow models to update beliefs as new information becomes available (Bodnar et al., 2021). Ensemble methods can also be interpreted as a form of robustness as they reduce dependence on any single model (Rodriguez Dominguez et al., 2025).

2.5 Previous Studies

Recent research has emphasized the importance of dynamic allocation across trading strategies. Dynamically allocating capital across multiple liquid futures

leads to significantly improved risk-adjusted returns compared to static equal-weighted approaches. Their results show that the optimal allocation between strategies depends on prevailing market conditions (Schmid Wirth, 2021)

Specifically, absolute trend-following strategies perform better in low-correlation environments, while relative strategies become more effective when correlations are high. This highlights that no single strategy is universally optimal. Their study also shows that dynamic allocation improves both Sharpe ratios and drawdown protection, while equal-weighted combinations often result in suboptimal performance.

(Bodnar et al., 2021) also highlights the challenges associated with estimation error in portfolio optimization. They show that parameter uncertainty introduces significant uncertainty in out-of-sample performance. Similarly, (DeMiguel Nogales, 2009) demonstrates that traditional covariance estimators are sensitive to extreme observations and that robust estimators can improve stability.

Additional approaches, such as shrinkage and regularization, have been proposed to reduce estimation risk and improve weight dispersion, especially in high-dimensional settings. Furthermore, (Bodnar et al., 2021) propose a volatility-sensitive Bayesian framework for estimating risk measures. It shows that models that adapt to changing volatility provide improved performance during turbulent periods.

Recent machine learning approaches such as deep reinforcement learning have also be explored for portfolio allocation (Hedlund, 2024). While these models can achieve strong performance, they remain sensitive to model design and do not fully eliminate robustness issues.

Finally, research on ensemble learning highlights that combining multiple predictive models can improve diversification and out-of-sample performance (Rodriguez Dominguez et al., 2025). This suggests that diversification can arise from both asset allocation and combining different predictive signals.

2.6 Research Gap

The literature highlights several key challenges in financial prediction and portfolio construction. First, financial markets are characterized by noise, non-stationarity and time-varying conditions. This means that no single strategy performs consistently well across all environments. Second, traditional portfolio optimization methods are highly sensitive to estimation error and parameter uncertainty. This often results in unstable weights and poor out-of-sample performance. Third, static allocation approaches provide robustness but fail to adapt to changing market dynamics.

At the same time, previous research suggests that both dynamic and model combination can improve performance. However, there remains a gap in understanding how multiple trading strategies can be combined using a simple data driven and adaptive mechanism.

This study addresses this gap by proposing an ensemble framework that assigns weights for trading strategies based on their recent performance with the aim of improving risk-adjusted returns in a dynamic market environment.

3 Theoretical Background

This chapter presents the theoretical foundations of the study, including momentum-based trading, diversification, ensemble methods, and risk-adjusted performance evaluation. It also discusses how previous research is applied and identifies relevant limitations.

3.1 Momentum and Trend-Following

Momentum describes the tendency for assets that have performed well in the past to continue delivering strong returns in the short term (Brooks, 2017). This effect is commonly exploited through trend-following strategies such as moving averages and breakout signals. In this study, momentum forms the basis for signal construction. Multiple indicators are used to capture different aspects of price behavior. However, momentum strategies may perform poorly in sideways or highly volatile markets. This motivates combining several signals instead of relying on a single approach.

3.2 Model Parameters and Design Choices

This section outlines the key parameter choices and design decisions underlying the implementation of both the Equal-Weighted and Adaptive Performance-Weighted models. The parameters were selected to balance responsiveness to market dynamics with robustness, while avoiding excessive model complexity and overfitting.

3.2.1 Signal construction and normalization

The model is based on five trend-following signals:

Time-Series Momentum (TSMOM)

$$TSMOM_t = sign\left(\frac{P_t}{P_{t-k}} - 1\right) \tag{1}$$

Where:

P_t is the asset price at time t.

P_{t-k} is the asset price k periods earlier.

k is the lookback window (e.g., $k = 20$ trading days).

$sign(\cdot)$ returns $+1$ if the argument is positive, -1 if negative, and 0 otherwise.

Moving Average Spread

$$MA_spread_t = \frac{MA_{short}(P_t)}{MA_{long}(P_t)} - 1 \tag{2}$$

$$MA_{short}(P_t) = \frac{1}{n_s} \sum_{i=0}^{n_s-1} P_{t-i}, \quad MA_{long}(P_t) = \frac{1}{n_l} \sum_{i=0}^{n_l-1} P_{t-i} \tag{3}$$

Where:

P_t is the asset price at time t.

n_s is the short-term moving average window (e.g., $n_s = 20$).

n_l is the long-term moving average window (e.g., $n_l = 60$).

MA_{short} and MA_{long} are simple moving averages.

Donchian Channel Breakout

$$Donchian_t = \frac{P_t - L_t}{H_t - L_t} \tag{4}$$

$$H_t = \max(P_{t-k+1}, \ldots, P_t), \quad L_t = \min(P_{t-k+1}, \ldots, P_t) \tag{5}$$

Where:

P_t is the asset price at time t.

H_t is the highest price observed over the past k periods.

L_t is the lowest price observed over the past k periods.

k is the lookback window (e.g., $k = 20$).

Regression Slope of Log-Prices

$$\log(P_{t-i}) = \alpha + \beta_t \cdot i + \varepsilon_i, \quad i = 0, \ldots, k-1 \tag{6}$$

$$Slope_t = \beta_t \tag{7}$$

$$Slope_t^{ann} = 252 \cdot \beta_t \tag{8}$$

Where:

P_t is the asset price at time t.

$\log(P_{t-i})$ is the natural logarithm of the price i periods before time t.

β_t is the estimated slope coefficient from the regression.

α is the intercept.

ε_i is the error term.

k is the regression window length (e.g., $k = 20$).

252 is the approximate number of trading days per year used for annualization.

Volatility-Adjusted Momentum

$$VolAdjMom_t = \frac{\log(P_t) - \log(P_{t-k})}{\sigma_t \sqrt{k}} \tag{9}$$

$$\sigma_t = std(\Delta \log P_{t-i}), \quad i = 1, \ldots, k \tag{10}$$

Where:

P_t is the asset price at time t.

$\log(P_t) - \log(P_{t-k})$ is the cumulative log-return over k periods.

σ_t is the standard deviation of daily log-returns over the past k periods.

$\Delta \log P_{t-i}$ denotes the log-return at time $t - i$. defined as the difference between the logarithm of the price at time $t-i$ and the logarithm of the price at the previous time step, $t - i - 1$

k is the lookback window (e.g., $k = 20$).

Rank-Based Normalization

$$z_t = 2 \cdot (rank_{t,k}(x_t) - 0.5) \tag{11}$$

Where:

x_t is the raw signal value at time t.

$rank_{t,k}(x_t)$ is the percentile rank of x_t within the past k observations.

k is the rolling window length (e.g., $k = 20$).

The transformation scales the signal to the interval $[-1, 1]$, where -1 corresponds to relatively low values and $+1$ to relatively high values. To ensure consistency across signals with different scales, all signals were normalized using a rank-based normalization. This was done using a 20-day rolling window to remove scale differences between the various indicators. Instead of standardizing the signals based on their mean and variance, each signal was transformed according to its relative position within its recent historical distribution. The value of each signal at time t was assigned a percentile rank within the rolling window. This results in a value between 0 and 1. This rank was then linearly scaled to the interval $[-1, 1]$ to ensure that all signals are directly comparable and centered around zero.

This approach ensures that extreme values are bounded and that the normalization is less sensitive to non-normal distributions and sudden spikes, making it suitable for financial time series.

Additionally, all signals are shifted by one period to eliminate look-ahead bias, ensuring that only information available at the time of decision making is used.

3.3 Portfolio Theory and Diversification

Portfolio theory highlights diversification as a means of reducing risk by combining assets that are not perfectly correlated (Markowitz, 1952). While traditionally applied to assets, diversification can also be applied to trading signals. In this study, diversification is implemented at the signal level by combining multiple indicators. This approach aims to improve robustness and reduce dependence on any single signal. Traditional optimization methods are not used due to their sensitivity to estimation errors and the unstable out-of-sample performance.

3.4 Ensemble Methods in Finance

Ensemble methods integrate multiple models or signals to enhance performance and robustness (Rodriguez Dominguez et al., 2025). In financial markets, this is useful since different strategies perform well under different conditions. This study applies an ensemble framework to combine multiple momentum signals. Two approaches are used: equal weighting and adaptive performance-based weighting.

3.5 Risk-Adjusted Performance Measures

Performance is evaluated using the risk-adjusted metrics Sharpe ratio, Sortino ratio, volatility, and maximum drawdown (Hedlund, 2024). These measures allow comparison of both return and risk, which is essential when evaluating portfolio strategies. However, these metrics have limitations, such as assumptions about return distributions.

3.6 Uncertainty and Model Robustness

Financial models are subject to uncertainty and changing market conditions. Parameter estimates based on historical data may not hold in the future (Lassance et al., 2023). To address this, adaptive and robust approaches are preferred (DeMiguel Nogales, 2009). In this study, the adaptive weighting model adjusts signal weights based on recent performance.

4 Method

This chapter describes the comparative framework used to evaluate the two ensemble strategies. The core of the study is a backtesting experiment where a baseline Equal-Weighted model as compared against the proposed Adaptive Performance-Weighted model. The methods were grounded in the theoretical frameworks described in the above section.

4.1 Procedure

The study followed a structured four-phase procedure to ensure that the backtesting experiment was both precise and reproducible. The procedure consisted of data preprocessing, signal generation, model implementation, and performance evaluation.

4.1.1 Phases

Phase 1: Data Acquisition and Pre-processing This phase involved preparing the dataset for analysis.

Phase 2: Signal Generation and Normalization In this phase, the signals used in the model were generated and standardized.

Phase 3: Model Implementation In this phase, the two weighting strategies were defined and applied within the framework to enable a comparative analysis.

Phase 4: Performance Evaluation In this phase, the outputs of the two approaches were compared in Excel.

4.2 Data collection

The proposed models were implemented in Python, using pandas and NumPy libraries for data manipulation and numerical calculations. Data retrieval was automated through the yfinance library, ensuring consistent access to adjusted historical prices. This technical setup enabled the processing of the multi-asset universe across the 9-year observation period.

The first step in the study was data collection and the selection of the instruments. Data retrieved via the Yahoo Finance API using the yfinance library. The investigated sample consisted of a diversified universe of 65 unique Exchange-traded Funds(ETFs) representing asset classes such as equities, fixed income, currencies, and commodities. The selected instruments are generally highly liquid. Although some niche and commodity focus instruments exhibit relatively lower liquidity. The purpose was to analyze a broad multi-asset universe to capture the premise that ”momentum is everywhere”. First, highly liquid ETFs were selected to ensure that the results reflect tradable market conditions and to minimize price discrepancies. Second, the instruments were chosen to represent performance across different market regimes. Third, the selection was limited to instruments with consistent daily price data available for the entire observation period. Furthermore, by including multiple asset classes, the analysis enables an evaluation of strategy performance under varying market conditions.

The data used consisted of daily observations of adjusted closing prices to account for the effects of dividends and stock splits. The observation period spanned from January 2017 to December 2025. A complete list of the included funds can be found in Appendix A. The choice of data parameters was guided by the need to ensure both statistical robustness and practical relevance.

First, adjusted closing prices were used instead of raw prices to accurately capture the total return of the assets. As the study spans multiple years, raw prices would be misleading due to dividends and stock splits. These reduce prices without affecting investor value. By using adjusted data, the momentum signals more accurately reflect the true economic performance of the instruments.

Second, daily observations were chosen to provide sufficient detail for the technical indicators. Signals such as the 20-day Moving Average and the Donchian channel require high-frequency data to remain reactive to short-term market shifts. Using lower-frequency data, such as monthly prices, would have resulted in significant ”lag”, making the adaptive weighting in the APW model less effective in volatile regimes.

Finally, the observation period (2017-2025) was selected to include a wide variety of market cycles, including periods of recovery, high inflation, and varying

volatility. A 9-year window provided a large enough sample size to ensure that the results were not a product of luck or a specific market environment. Furthermore, this long duration was necessary to accommodate the model's 20-day "warm-up" period for normalization and the performance-based lookback window.

4.3 Technical Parameters and Experimental Framework

A fixed set of technical parameters was established to ensure a fair comparison analysis between the Equal-weighted (EW) and the Adaptive Performance-Weighted (APW) models. These parameters defined the lookback windows for signal generation, the normalization process, and the decision-making logic of the ensemble.

The experiment used daily price observations covering January 2017 to December 2025. This 9-year period was chosen to expose the models to various market regimes, which included the COVID-19 pandemic and following inflationary environments. Using a consistent timeframe for all 65 instruments was essential to maintain statistical consistency across multi-asset universe.

The five indicators within the ensemble used standardized lookback periods to capture trend dynamics. A 20-day short term, 60-day long-term moving average was implemented for the trend-following analysis. For Donchian breakouts, Time-series Momentum, and annualized log-price regression slope, a consistent 20-day window was used to maintain sensitivity to monthly price changes.

A rank-based normalization was applied using a 20-day rolling window to ensure consistency across signals and reduce sensitivity to outliers. Each signal was transformed into its percentile rank within the window, yielding values in $[0, 1]$, which were then linearly rescaled to $[-1, 1]$:

$$z_t = 2 \cdot (rank_{t,20}(x_t) - 0.5) \tag{12}$$

Where x_t denotes the raw signal at time t, and $rank_{t,20}(x_t)$ its percentile rank over the past 20 observations.

The normalization step was essential for the ensembles' aggregation logic.

Furthermore, it served to mitigate the impact of extreme outliers and ensure that a single volatile instrument did not disproportionately influence the final decision of the Adaptive Performance-Weighted and Equal-Weighted models.

The Adaptive performance-Weighted (APW) model uses specific windows to dynamically adjust signal weights based on recent performance. A 120-day lookback period was used to assess the historical accuracy of each signal, which was then validated against a 10-day forward prediction horizon.

Finally, a three-state decision framework was implemented for trade execution. A trade recommendation was only triggered if the aggregated ensemble signal reached a strength threshold of +- 0.3. This specific threshold was selected to ensure that trade execution only occurred when there was a significant conviction among the underlying indicators. A final signal above 0.3 resulted in a ”BUY” recommendation, while a signal below -0.3 triggered a ”SELL” recommendation. Any signal strength within the +-0.3 range was classified as ”NEUTRAL”, which represents a lack of consensus among the underlying indicators. This ”dead zone” was essential to prevent overtrading in sideways markets and to ensure that the models only reacted to trend signals where sufficient consensus had been reached.

4.3.1 Signal construction and normalization

The models use five trend-following indicators: time-series momentum, moving average spread, Donchian channel breakout, regression slope of log-prices, and volatility-adjusted momentum. These indicators are described theoretically in Chapter 3.

In the implementation, all signals were computed using daily adjusted closing prices. A 120-day lookback window was used for time-series momentum, Donchian breakout, regression slope, and volatility-adjusted momentum. For the moving average spread, a 20-day moving average and a 60-day moving average were used.

To ensure consistency across indicators, each raw signal was transformed using a 20-day rolling rank-based normalization and rescaled to the interval $[-1, 1]$. All

signals were then shifted by one trading day to avoid look-ahead bias. This was done to ensure that only information available at the time of portfolio formation was used.

4.3.2 Prediction Horizon and Evaluation Window

The prediction horizon is set to ten trading days, meaning that each signal is evaluated based on its ability to correctly predict the direction of price movement over the following two weeks. This choice reflects a short-term trading perspective while avoiding excessive noise associated with daily predictions. Alternative horizons were also tested, and the results are discussed in the sensitivity analysis.

To determine the relative performance of each signal, a rolling lookback window of 120 trading days is used. Within this window, each signal is evaluated based on the number of correct directional predictions. This design introduces flexibility, allowing the model to adjust to changing market conditions by pointing out signals that have recently performed well. Alternative lookback specifications were examined as part of a sensitivity analysis.

4.3.3 Weighting schemes

Two different ensemble strategies are implemented:

Equal-Weighted Model All signals are assigned identical weights, implying that each signal contributes equally to the final decision. This serves as a baseline model and reflects a neutral assumption regarding signal quality.

Adaptive Performance-Weighted Model Signals are ranked based on their predictive accuracy over the lookback window. Weights are then assigned in decreasing other according to rank, with the best-performing signal receiving the highest weight and subsequent signals receiving gradually lower weights. In the implemented version, weights decrease linearly(e.g., 1.0, 0.8. 0.6, etc).

This approach allows the model to dynamically prioritize signals that have recently had good predictive strength. At the same time maintaining, diversification

across multiple strategies. The decreasing weighting scheme prevents excessive concentration in a single signal to reduce the risk of overfitting to short-term noise.

The final trading signal is constructed as a weighted combination of the individual signal directions. Specifically, the ensemble signal is defined as:

$$S_t = \frac{\sum_{i=1}^{N} w_i \cdot sign(z_{i,t})}{\sum_{i=1}^{N} w_i} \tag{13}$$

Where:

w_i represents the weight assigned to signal i, and $sign(z_{i,t})$ denotes the directional prediction of signal i at time t.

In the Equal-Weighted model, all weights are identical $w_i = 1$. While in the Adaptive Performance-Weighted model, the weights depend on the recent predictive performance of each signal. This formulation ensures that the final signal reflects both the direction and the relative importance of each component strategy.

4.3.4 Decision Thresholds

The final trading signal is computed as the normalized weighted sum of all individual signal directions. To translate this continuous signal into actionable decisions, fixed thresholds are applied:

- Buy if $S_t > 0.3$
- Sell if $S_t < -0.3$
- Otherwise, remain neutral

To convert the continuous ensemble signal into actionable trading decisions, a threshold-based rule is applied. This decision rule requires a minimum level of agreement among the underlying signals before initiating a position. The threshold ± 0.3 was selected to balance sensitivity and robustness. This ensures that trades are only executed when there is a sufficiently strong consensus among the signals.

4.3.5 Translation of Signals into Portfolio Positions

After the individual signals were aggregated into a final ensemble signal. The signal was translated into a portfolio position using the predefined decision thresholds. If the final signal was above 0.3, the asset received a long position. If the signal was below -0.3, the asset received a sell recommendation. Signals between -0.3 and 0.3 were classified as neutral, meaning no active position was taken.

At each balancing date, the model generated recommendations for all ETFs in the asset universe. The assets with the strongest positive signals were selected for the portfolio. Portfolio capital was allocated across the selected assets according to the output of the model. In the Equal-Weighted model, selected assets were assigned equal portfolio weights. In the Adaptive Performance-Weighted model, allocation was influenced by the recent performance-based signal weights. This gives greater importance to the signals that had shown stronger predictive accuracy during lookback period.

This procedure ensured that the trading signals were not only used as directional indicators. But they were directly translated into portfolio positions within the backtesting framework.

4.3.6 Rebalancing Frequency

The model operates on daily data but applies a monthly balancing schedule in the backtesting procedure. This design reduces transaction frequency and associated trading costs while maintaining sufficient responsiveness to evolving markek conditions.

A lower balancing frequency also mitigates the risk of overreacting to short-term fluctuations. This is relevant when combining multiple noisy signals.

4.3.7 Design considerations

Overall, the parameter choices reflect a balance between responsiveness and stability. Short-term signals and evaluation windows allow the model to react to changing market regimes, while normalization, thresholding and moderate

balancing frequency contribute to robustness.

Importantly, all parameters are predefined rather than optimized through extensive tuning procedures. This design choice reduces the risk of overfitting and ensures that the comparison between the Equal-Weighted, and Adaptive models remains fair and methodologically sound.

4.4 Theoretical Framework

This chapter presents the theoretical foundations underlying the proposed ensemble framework. The study builds on three key concepts: momentum-based trading, diversification across multiple signals, and risk-adjusted performance evaluation.

4.4.1 Momentum and Trend-Following

Momentum refers to the tendency of assets that have performed well in the recent past to continue performing well. While assets that have performed poorly tend to continue underperforming over a subsequent period. In practice, momentum-based strategies attempt to exploit persistent price trends by using historical price information as an indicator of future directional movement.

In a broad sense, momentum can be understood as a continuation effect in returns. Rather than assuming that prices instantaneously reflect all available information, momentum-based approaches assume that market adjustments may occur gradually. This creates temporary persistence in price movements, which can be captured through trend-following rules such as moving averages, breakout indicators, and return-based momentum measures.

A simple representation of momentum over a lookback window k can be written as:

$$M_t = \frac{P_t - P_{t-k}}{P_{t-k}} \tag{14}$$

Where:

P_t denotes the asset price at time t, and P_{t-k} denotes the price k periods earlier.

A positive value indicates upward momentum, while a negative value indicates downward momentum.

In this study, momentum is not represented by a single indicator, but rather by several related trend-following signals. Although these signals differ in construction, they are all based on the same underlying principle that historical price dynamics may contain information about short-term future direction. The purpose of combining several momentum-related signals is to capture them from different analytical perspectives.

Momentum is relevant for this study because it provides the foundation for the signal generation process. The proposed ensemble models do not attempt to forecast prices using fundamental valuation or macroeconomic variables. But instead, rely on price-based indicators. For that reason, an understanding of momentum is necessary to interpret both the model design and the empirical results.

To ensure alignment across signals with different scales, it is necessary to standardize them. This is commonly done using statistical normalization techniques. This adjusts signals relative to their historical distribution. Normalization ensures that no single signal dominates the model purely due to differences in scale or volatility.

4.4.2 Diversification in Multi-Signal Portfolios

Diversification is a principle in portfolio theory and refers to the idea that combining multiple asset, strategies, or signals can reduce total risk when the components are not perfectly correlated. The basic intuition is that different return sources do not perform equally well under all market conditions. As a result, combining them may smooth the overall performance profile and reduce the impact of weakness in any single component.

In traditional portfolio theory, diversification reduces idiosyncratic risk by spreading exposure across assets whose returns are imperfectly correlated. The same logic can be extended from assets to trading signals. If different signals react differently to market environments, combining them may lead to a more stable

decision process than relying on a signal alone.

It can be illustrated through the standard two-asset portfolio variance formula:

$$\sigma_p^2 = w_1^2\sigma_1^2 + w_2^2\sigma_2^2 + 2w_1w_2\sigma_1\sigma_2\rho_{12} \tag{15}$$

Where:

w_1 and w_2 are the portfolio weights, σ_1 and σ_2 are the standard deviations of the two assets, and ρ_{12} is the correlation between the two components. The formula shows that overall variance depends on both the volatility and the correlation between them. When correlation is below one, diversification can reduce total portfolio risk.

In the context of this study, diversification is relevant at the signal level. The ensemble framework combines multiple trend-related signals, each constructed in a different way. Even though all signals are linked to momentum, they capture different aspects of market behavior. For example, some signals focus on recent returns, other on price breakouts, and other on the slope or persistence of price development. Because these signals may not react identically at all times, combining them can improve robustness.

This is important because no single signal is consistently superior across all regimes. A strategy that performs well in trending markets may struggle in sideways or highly volatile period. By aggregating several signals, the ensemble approach seeks to reduce dependence on any one signal and thereby improve stability over time. The distinction between the Equal-Weighted and Adaptive Performance-Weighted models is therefore not whether diversification is used, but how the contribution of each signal is determined within the diversified framework.

4.4.3 Risk-Adjusted Performance and the Sharpe Ratio

Evaluating an investment strategy based solely on return can be misleading, since higher returns may simply be a result of taking greater risk. For this reason,

financial performance is often assessed using risk-adjusted measures that relate returns to the volatility or downside profile of those returns. One of the most widely used measures in this context is the Sharpe ratio.

The Sharpe ratio measures the excess return of a portfolio relative to the risk-free rate per unit of total risk. Where risk is represented by the standard deviation of the returns, and the risk-free rate is set to 3%. It is written as:

$$SharpeRatio = \frac{R_p - R_f}{\sigma_p} \tag{16}$$

Where:

R_p is the average portfolio return, R_f is the risk-free rate, and σ_p is the standard deviation of portfolio returns.

The intuition behind the Sharpe ratio is straightforward. A strategy is considered more attractive if it delivers more excess return for each unit of risk taken. A higher Sharpe ratio, therefore indicates better risk-adjusted performance. This makes the metric useful when comparing two strategies that may generate different levels of return and volatility.

In this study, the Sharpe ratio is relevant because the objective is not merely to determine whether adaptive weighting produces higher raw returns than equal weighting. But whether it improves performance on a risk-adjusted basis. Since the Adaptive Performance-Weighted model may alter both the return distribution and the volatility profile of the strategy, an evaluation framework based only on cumulative return would be incomplete. The Sharpe ratio helps capture whether any performance improvement is achieved efficiently rather than through increased exposure.

At the same time, the Sharpe ratio has limitations. Because it uses total volatility as the denominator, it treats upside and downside risk the same. For this reason, the Sharpe ratio is complemented by the Sortino ratio, maximum drawdown, and annualized volatility. Nevertheless, the Sharpe ratio remains a standard and important benchmark for assessing whether a trading strategy delivers sufficient return relative to its overall risk.

4.5 Implementation and Execution

This section describes the implementation of the framework and the execution. The implementation follows a structured pipeline, where data is collected, processed, transformed into signals, and then used for portfolio decision making.

4.5.1 Practical implementation

The models were implemented in Python using Pandas and NumPy for data handling and numerical computations. Yahoo finance(yfinance) was used for data retrieval.

Historical daily price data was downloaded for a predefined set of 65 ETFs using an automated pipeline. The data was stored locally to ensure reproducibility and reduce API calls.

All datasets were aligned to the same time index and sorted chronologically. Missing values were handled, and adjusted closing prices were used to reflect total return. To prevent look-ahead bias, all the signals were shifted by one time step. This was done to ensure that only information available at the time of decision making was used.

The implementation was structured as a modular pipeline consisting of separate functions for data collection, signal generation, normalization, and model evaluation.

4.5.2 Data Processing and Signal Execution

For each ETF, a set of five trend-following signals was computed using rolling windows applied to daily price data. These signals were time-series momentum, moving average spread, Donchian breakout, regression slope and volatility-adjusted momentum.

Each signal was calculated iteratively across the full dataset. A normalization step was applied using rolling transformation to ensure consistency across signals with different scales.

The signal generation and normalization process was applied iteratively for each asset and each time step. Rolling windows were used for all signal calculations, and all signals were shifted by one period to prevent look-ahead bias. This ensured that only information available at the time of decision making was used in the model.

4.5.3 Model Execution

The models were executed using a rolling backtesting framework. For each time step, signals were calculated for all assets, and a final recommendation was generated.

Two models were implemented and evaluated. The first one which is the base, is an Equal-Weighted model and the second is an Adaptive Performance-Weighted model. In both cases, signals were combined into a single aggregated signal. These were used to generate trading decisions.

In the adaptive model, signal performance was evaluated over a rolling lookback window. Each signal's predictive accuracy was evaluated by comparing its directional prediction to the actual future price movement over a predefined horizon. The number of correct predictions was calculated over a rolling lookback window, and the signals were ranked based on this accuracy measure.

Weights were assigned according to this ranking. The higher weights were given to signals with better recent performance. The final signal was computed as a weighted average of all individual signal directions.

The implementation was applied across all assets in an iterative loop. Each ETF was processed individually. Also, the model was executed repeatedly for multiple end dates to simulate a rolling monthly portfolio construction process. For each run, recommendations were generated and ranked based on signal strength. This allows the selection of the top-performing assets.

4.5.4 Application of Methods

The theoretical framework was directly translated into the implementation. Momentum indicators were used to generate the signals from the historical price

data.

The ensemble approach was implemented by combining multiple signals into a single decision. All signals contributed equally in the Equal-Weighted model. This is to reflect a static diversification approach. On the other hand, we have the Adaptive model which dynamically adjusts signal weights based on recent predictive performance.

This implementation reflects the concept that no single strategy consistently outperforms across all market conditions. The model adapts to changing environments by dynamically allocating weights, while maintaining diversification across signals.

4.5.5 Challenges and Limitations

Several practical challenges occur during the implementation. We had some Data issues, that included inconsistencies in the downloaded datasets and some missing values. These problems required extra preprocessing to ensure consistency.

Another limitation is how the signal performance is evaluated in the adaptive model. The weights are based on the past predictive accuracy. But as we mentioned earlier, the financial markets change over time. Meaning that signals that performed well in one period might perform badly in the future. This means that the model sometimes assigns high weights to signals that are no longer effective.

Our weighting approach simplifies the difference between signals. Signals are ranked based on their performance. But the method does not fully capture how large the differences actually are. With this, two signals with similar performance can still receive different weights, and that will affect the final decision.

Another one is that signals are based on similar price information. This means that they may correlate a lot and not be fully independent. This takes the diversification benefit of combining multiple signals to lower the correlation.

The model also uses fixed thresholds to generate buy and sell decisions. These

thresholds do not change with the market conditions. Even if the trends and volatility change over time. So they limit the model's ability to adapt to different market environments.

Another issue was the model's sensitivity to parameter choices. The lookback windows and normalization methods all affect the outcome of the model. These parameters can have a strong impact on how the model behaves.

Finally, the backtesting assumed frictionless markets. Transaction costs and liquidity were not included. This may limit the real-world practical value of the results.

5 Results

5.1 Portfolio Performance Overview

5.1.1 Annual Performance Comparison

Figure 5.1 presents the annual returns of the Equal-Weighted (EW) and Adaptive Performance-Weighted (APW) portfolios over the full observation period. It provides an overall picture of how the two portfolio construction approaches evolve over time when returns are compounded across the sample.

The cumulative return plot shows which strategy achieves the highest final value and how performance is generated throughout the period. The figure makes it possible to assess whether performance is consistent or concentrated to a limited number of sub-periods. Compared with the annual return table 5.1 alone, the cumulative returns provide a representation of the growth by both strategies.

As we can observe in Figure 5.1, the APW portfolio ends the period with a return above the EW portfolio. This indicates that the adaptive weighting has been successful in improving capital growth over the long run. The result suggests that assigning greater weight to recently better performing signals allows the model to adjust more effectively to changing market conditions than a static equal-weighted structure. The periods in which the APW curve diverges upward from the EW curve may be interpreted as intervals during which signal selection and dynamic weighting added value relative to the benchmark approach.

At the same time, the cumulative return figure also reveals whether improved total return comes at the cost of greater volatility. The APW portfolio shows sharper fluctuations or more irregular growth. This indicates that the strategy may be more sensitive to shifts in the market conditions. The EW portfolio follows a smoother path. This support that EW provides is a more stable but less responsive allocation method.

An important aspect of the figure is the shape of the return trajectories during adverse market periods. Both portfolios decline simultaneously. This suggests that the two strategies remain exposed to common market dynamics despite

differences in internal weighting methodology. However, the APW portfolio recovers faster after drawdowns, this mean that the APW contributes to improved resilience by reallocating toward signals that regain predictive strength more quickly.

The results show that the Adaptive Performance-Weighted portfolio generally achieves higher returns in several periods, particularly toward the later years of the sample. The APW portfolio reached returns of 11.05% in the final year. This is outperforming the Equal-Weighted portfolio, which achieves 4.06% in the final year. Thus, the APW is able to better take advantage of market conditions by allocating more weights to recently successful signals.

However, the advantage in performance is not consistent across all years. Both portfolios have periods of negative returns, with the most drawdowns occurring around the pandemic. The APW portfolio returns a minimum annual return of -9,42% while the EW portfolio reaches -9,72% in the year 2021. Thus, both strategies are exposed to similar downside risk during adverse market conditions. This aligns with the expectation that all momentum based strategies may struggle during volatile or non-trending environments.

Overall, Table 5.1 provides a view of the return generation across the sample period. It serves as the main visual summary of the long-term portfolio performance and forms the basis for the more detailed comparisons presented in the following figures.

Table 5.1: Annual and cumulative returns for adaptive performance-weighted (APW) and equal-weighted (EW) portfolios.

Method	2017	2018	2019	2020	2021	2022	2023	2024	2025	Cumulative
APW	5.78%	-2.50%	3.59%	-0.44%	-9.42%	0.19%	5.80%	0.82%	11.05%	14.88%
EW	4.87%	-3.93%	-0.53%	8.44%	-9.72%	-2.93%	1.62%	2.79%	4.06%	4.68%

Figure 5.1 below shows the annual returns of the EW and APW portfolios across the sample period. This figure isolates performance year by year and makes a more detailed comparison of how the strategies behave in different market environments.

The year-by-year perspective is important in the context of this study. Since the purpose of the APW model is to adapt to changing market conditions

by assigning greater weight to signals that have recently performed stronger predictive performance. A consistent pattern of higher returns for the APW supports the hypothesis. It suggests that dynamic signal weighting can improve portfolio outcomes relative to a static benchmark.

The reported annual results suggest that the APW portfolio achieves stronger performance over several years. This difference indicates that the adaptive portfolio was more effective at taking advantage of favorable market conditions during those periods.

At the same time, the figure also shows that the performance advantage of the APW strategy is not uniform across all years. There are years in which both portfolios produce weak or negative returns. There are also some periods where the EW portfolio performs a bit better. For example, the EW strategy records stronger returns in one of the earlier positive years, while the APW portfolio underperforms in certain intermediate periods. This indicates that the adaptive approach does not dominate in every environment and that the model's responsiveness may sometimes lead to weaker outcomes when recent signal performance is less informative about future returns.

Figure 5.1 highlights that both portfolios experience significant negative years, with minimum annual returns of -9,42% for APW and -9,72%for EW. The similar downside results suggest that the adaptive weighting does not prevent losses when market conditions are unfavorable. Instead, it seems to have a stronger impact during positive periods than during downturns. This is important, as it indicates that the main strength of the APW model may be its ability to capture gains in favorable markets rather than to fully protect against losses.

From Figure 5.2, we observe the difference in returns between the two strategies. The APW portfolio appears to exhibit a wider range of yearly outcomes, combining stronger positive years with some negative periods. In contrast, the EW portfolio appears more stable in its yearly profile, but this stability is associated with lower upside potential.

Taken together, the figure below suggests that the APW strategy improves return generation in several important subperiods. But this improvement is not constant

across all years. The results therefore, support a mixed interpretation. The adaptive weighting can enhance performance under certain market regimes, but the benefits depend on the extent to which recent signal performance remains informative in the near future.

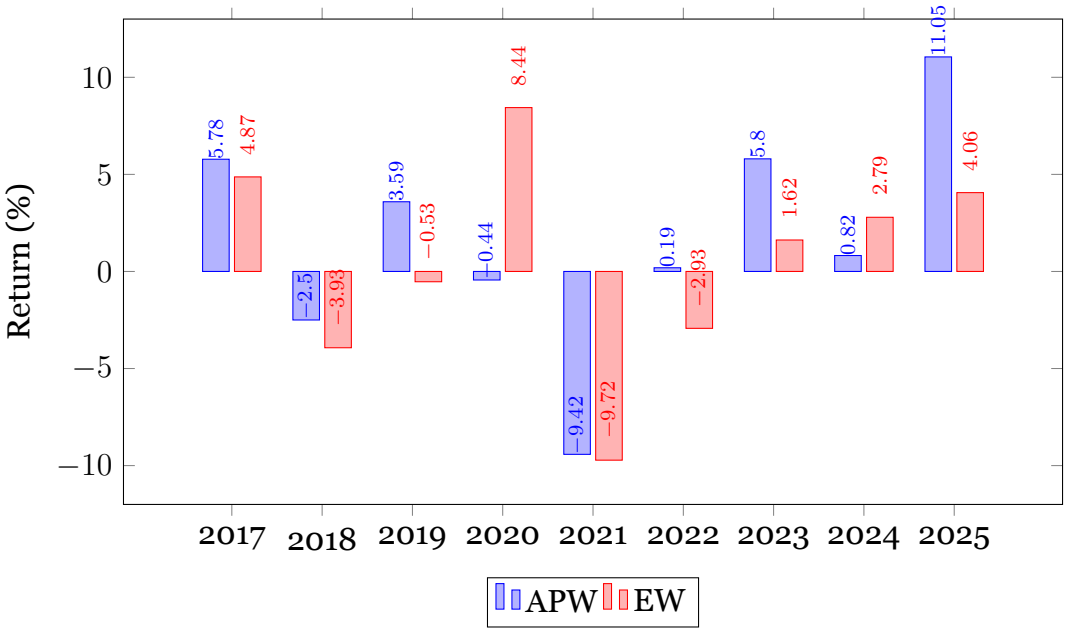


Figure 5.1: Annual returns for APW and EW portfolios from 2017 to 2025.

5.1.2 Monthly Performance Analysis

Figure 5.2 presents the monthly returns of the Equal-Weighed (EW) and Adaptive Performance-Weighted (APW) returns over the full period from 2017 to 2025.

The results show that both portfolios have variation in monthly returns. This reflects the inherent volatility of financial markets. However, a clear difference emerges in the dispersion of the returns between the two strategies. The APW portfolio shows more fluctuations with both stronger positive and more negative monthly outcomes compared to the EW portfolio.

During periods of strong market movement, the APW portfolio achieves higher returns with approximately 5.25% in 2020 and 6.50% in 2023. These peaks exceed the corresponding EW returns in the same period. So the adaptive weighting is able to allocate more effectively to signals that perform well in

trending environments. Similarly, in 2024 and 2025, the APW portfolio records several strong positive months, including returns above 6%. This highlights its ability to capitalize on favorable short-term opportunities.

At the same time, the APW portfolio experiences more pronounced negative outcomes. It has a monthly loss of approximately -7.76% in 2020 and -6.91% in 2022. These are slightly larger than those for the EW portfolio. So the increased responsiveness of the adaptive model also exposes it to greater short-term down-side risk. This has a higher chance to happen during abrupt market reversals or period of signal noise.

On the other hand, the EW portfolio has a more stable monthly return profile. While it also experiences both gains and losses, the size of these fluctuations is generally smaller. Thereby this reduces the impact of individual signal misclassifications. This stability comes at the cost of reduced ability to use strong trends, as reflected in the smaller size of positive monthly returns.

Another important observation is that both portfolios tend to experience positive and negative months simultaneously. This indicates that they are influenced by the same market conditions. So the difference between them primarily is in how strongly they react rather than in fundamentally different directional exposure. In other words, that adaptive model increases the effects of market movements instead of changing the overall direction of returns.

The Adaptive Performance-Weighted portfolio improves responsiveness to short-term market dynamics. This results in higher return potential during favorable periods. But it comes with greater volatility and exposure to short-term losses. On the other hand, Equal-Weighted portfolio provides a more stable but less dynamic performance profile.

Table 5.2 provides a compact overview of the monthly returns for the APW and the EW portfolios across the full sample period from 2017 to 2025. Each cell reports the corresponding monthly return in percentage terms. This allows for a detailed comparison of short-term performance between the two strategies.

The table highlights that both portfolios are faced with the same market conditions. Returns on positive and negative months often occur simultaneously.

Both strategies are driven by similar directional signals, while differences in performance arise primarily from the weighting.

Second, the APW portfolio has more variation in returns. This is obvious in months with strong performance, such as March 2020(2.27% vs. 6.51%) and December 2023 (6.50% vs. 5.06%). However, the table also shows that this increased responsiveness leads to more negative outcomes in certain periods, including July (-7.76% vs. -6.65%) and April 2022 (-6.91% vs. -5.47%). This reinforces the observation that the adaptive model amplifies both upside and downside movements.

Third, the EW portfolio demonstrates a more stable return pattern, with fewer extreme monthly values. Even if this reduces exposure to large losses, it also limits the ability to use strong trends. This trade-off between stability and responsiveness is consistently observed throughout the table.

Additionally, the table reveals that performance differences between the two strategies vary over time and across market regimes. In some years, such as 2024 and 2025, the APW portfolio frequently outperforms the EW portfolio in individual months. This suggests that the adaptive model is effective in more structured or trending environments. But during more turbulent periods like 2021 and 2022, both portfolios have negative months.

Overall, Table 5.2 supports the findings about the APW portfolio. This results in higher volatility and sometimes outperformance. This while the EW portfolio maintains a more stable but less dynamic returns profile.

Table 5.2: Compact monthly overview of returns for the APW and EW portfolios. Each cell reports APW/EW returns in percent.

Year	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
2017	1.15/1.30	1.00/1.00	-0.04/-0.32	0.78/1.52	0.69/0.30	-0.34/-1.40	1.30/0.55	0.56/2.07	-1.08/-1.21	-1.27/-0.24	0.44/0.03	2.50/1.23
2018	2.71/3.75	-3.81/-2.30	-0.31/0.25	-0.03/1.34	1.62/-0.71	0.05/0.16	-0.76/-1.16	0.29/0.00	-0.08/-0.14	0.20/-0.94	1.15/0.85	-3.37/-4.86
2019	0.84/0.61	1.01/0.33	1.06/0.73	-0.07/-0.51	-2.16/-0.07	-0.43/0.54	-1.61/-0.66	2.42/-1.83	0.60/0.83	0.27/0.22	2.58/1.48	-0.86/-2.14
2020	-2.00/-0.94	1.67/2.52	5.27/6.51	1.06/1.46	2.05/5.28	1.75/1.37	-7.76/-6.65	-0.35/0.08	0.11/1.45	-0.89/-0.84	-4.95/-4.87	4.35/3.56
2021	-0.27/1.19	-0.89/-0.84	3.28/0.58	-1.54/-3.79	1.42/0.75	-0.42/0.44	-1.47/-0.01	0.27/-0.95	-0.61/2.21	-3.51/-6.37	-2.70/-1.81	-3.21/-1.27
2022	-3.21/-2.40	1.42/0.63	2.35/3.19	-6.91/-5.47	2.44/-1.06	-0.59/-4.50	-0.96/-5.48	-3.41/-2.47	2.44/3.40	3.63/3.53	5.45/7.93	-1.77/0.68
2023	-1.85/-2.19	-6.09/-6.34	-1.05/-0.55	1.22/0.51	0.63/0.82	3.00/1.52	3.14/1.84	-3.30/-4.53	1.08/3.64	2.78/1.79	0.20/0.62	6.50/5.06
2024	-1.56/-2.25	-0.86/-0.35	2.49/1.15	2.79/0.93	-2.72/0.61	-1.07/0.25	1.34/1.12	1.13/0.54	6.24/3.24	-1.24/-0.98	-1.85/0.66	-3.46/-2.04
2025	-3.85/-4.13	0.43/1.47	2.45/2.29	2.37/-1.72	4.70/3.89	-1.20/1.78	0.19/0.40	-2.20/-4.56	3.52/3.49	3.08/1.55	0.77/-0.30	0.61/0.23

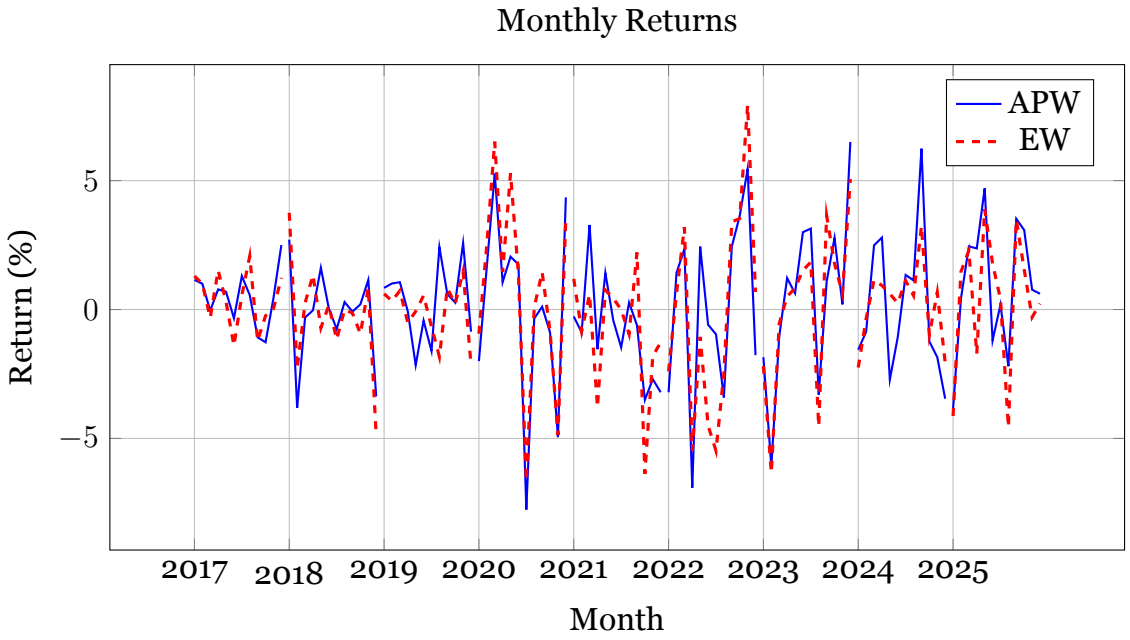


Figure 5.2: Monthly returns for the APW and EW portfolios from 2017 to 2025.

5.2 Risk-Adjusted Performance

Tables 5.3 and 5.4 below presents a comparison of the key risk-adjusted performance metrics for the Equal-Weighted(EW) and Adaptive Performance-Weighted(APW) portfolios over the full sample period. By combining return, volatility, Sharpe ratio, Sortino ratio and maximum drawdown in a single visualization, the figure provides a comprehensive summary of the trade-offs between return generation and risk exposure for the two strategies.

From a return perspective, the APW portfolio demonstrates stronger performance in several periods. For example, in 2025, the APW portfolio achieves a return of 11.05% compared to 4.06% for the EW portfolio. Similarly, in 2023 with the APW strategy had a return of 5.80% compared to 1.65% achieved by the EW portfolio. These results indicate that the adaptive weighting is able to enhance return generation during favorable market conditions.

However, this improved return potential must be evaluated in relation to risk. The volatility measure show that both portfolios experience variation in returns. It is very high during the turbulent period of 2020 and 2022. In 2020, volatility reaches approximately 12.67% for APW and 13.14% for EW, while in 2022, both portfolios show volatility above 12%. So neither strategy is able to avoid

exposure to market-wide fluctuations, and the adaptive approach does not reduce risk.

The Sharpe ratio provides further insight into the efficiency of return generation. The results show that both portfolios struggle to achieve positive Sharpe ratios across the whole sample period. Both strategies show negative Sharpe values in 2021 with APW having -1.86 and EW -1.56. This reflects the poor performance relative to risk during that year. However, in later periods, such as 2025, the APW portfolio achieves a higher Sharpe ratio of 0.94 compared to 0.11 for the EW portfolio. This suggests that the adaptive strategy is capable of improving risk-adjusted performance under favorable conditions even if overall efficiency remains inconsistent.

A similar pattern is observed for the Sortino ratio, which focuses specifically on the downside risk. The APW portfolio demonstrates higher Sortino values in strong performance years such as 2025(1-74 VS 0.2). This indicates improved compensation for downside risk. However, during an adverse period such as 2021, both portfolios have highly negative Sortino values with APW reaching -2.87. This highlights that neither strategy was able to effectively mitigate downside exposure during unfavorable market regimes.

The maximum drawdown further reinforces this conclusion. Both portfolios have large drawdowns during periods of market stress. In 2021, the APW portfolio fell by -10.03% while the EW portfolio falls -9.45%. In 2022, the EW portfolio experienced an even larger drawdown of -18.98% compared to -6.91% for APW. This could mean that the adaptive approach may provide some protection in certain periods. However, this advantage is not consistent across all years.

Overall, Table 5.4 shows that the Adaptive Performance-Weighted portfolio improves return potential and, in some cases, risk-adjusted performance. However, both portfolios have considerable fluctuations and periods of weak performance. So the adaptive weighting enhances performance in specific market conditions but does not fundamentally eliminate exposure to risk.

Table 5.3: Annual return and volatility for APW and EW portfolios, 2017–2025.

Method	Metric	2017	2018	2019	2020	2021	2022	2023	2024	2025
APW	Return	5.78%	-2.50%	3.59%	-0.44%	-9.42%	0.19%	5.80%	0.82%	11.05%
APW	Volatility	3.63%	6.40%	5.00%	12.69%	6.68%	12.20%	11.50%	9.60%	8.615%
EW	Return	4.87%	-3.93%	-0.53%	8.44%	-9.72%	-2.93%	1.62%	2.79%	4.06%
EW	Volatility	3.74%	7.17%	3.74%	13.14%	8.14%	14.43%	11.18%	5.19%	9.330%

Table 5.4: Annual Sharpe ratio, Sortino ratio, drawdown volatility, and maximum drawdown for APW and EW portfolios, 2017–2025.

Method	Metric	2017	2018	2019	2020	2021	2022	2023	2024	2025
APW	Sharpe	0.77	-0.86	0.12	-0.27	-1.86	-0.23	0.24	-0.23	0.935
APW	Sortino	1.37	-1.13	0.19	-0.32	-2.87	-0.35	0.36	-0.67	1.74
APW	Vol(Drawdown)	2.03%	4.87%	3.15%	10.71%	4.33%	8.01%	7.68%	3.28%	4.64%
APW	Max drawdown	-2.35%	-4.15%	-2.23%	-8.11%	-10.03%	-6.91%	-8.99%	-6.55%	-3.85%
EW	Sharpe	0.50	-0.97	-0.94	0.41	-1.56	-0.41	-0.12	-0.04	0.114
EW	Sortino	0.90	-1.07	-1.36	0.54	-1.45	-0.93	-0.16	-0.07	0.20
EW	Vol(Drawdown)	2.07%	6.47%	2.60%	10.06%	8.76%	6.39%	8.83%	3.11%	5.30%
EW	Max drawdown	-1.45%	-4.86%	-2.49%	-6.65%	-9.45%	-18.98%	-9.08%	-2.60%	-4.56%

5.2.1 Distribution of Portfolio Returns

Figure 5.3 illustrates the monthly return for the Equal-Weighted (EW) and Adaptive Performance-Weighted (APW) portfolios. The distribution reveals that both portfolios generate returns around small values with a high concentration of observations close to zero. Most months have moderate performance with few extreme outcomes. However, there is a difference between the two strategies in the dispersion of returns.

The APW portfolios shows a wider distribution with fatter tails on both the positive and negative sides(more extreme values, more like t-students distribution). The adaptive strategy produces both stronger gains and larger losses. For example, the APW portfolio includes extreme positive observations above 6%(e.g., 6.5%in 2023 and 6.24% in 2024). Also, significant negative outcomes such as -7.76% in 2020 and -6.91% in 2022. So the model have a higher level of variation and responsiveness to market conditions.

But we can see that the EW portfolio displays a more concentrated distribution. It has fewer extreme values and a narrower spread of returns. This means that the Equal-Weighted approach produces a more stable return profile. But this comes at the cost of reduced ability to capture strong market movements. Both strategies show a presence of negative outcomes. This helps explain the weak Sharpe and

Sortino ratios observed in the risk-adjusted analysis. The existence of several large negative observations means that both portfolios remain exposed to unfavorable market conditions despite diversification across signals.

Overall, Figure 5.3 confirms that the Adaptive Performance-Weighted portfolios increase return dispersion, leading to greater upside potential but also increased downside risk.

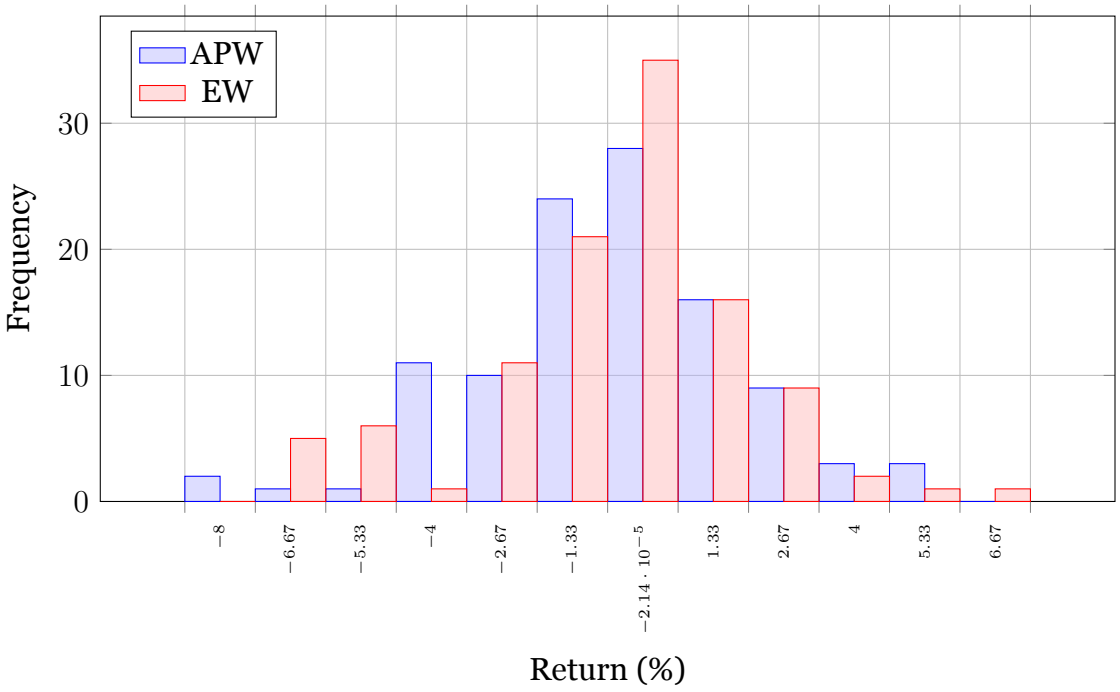


Figure 5.3: Histogram of monthly returns for APW and EW portfolios.

Taken together, the results provide a view of the trade-offs between risk and return and the risk for the two portfolios' construction approaches. While the APW portfolio demonstrates an improved ability to generate higher returns in certain periods, this improvement is not consistently reflected in stronger risk-adjusted performance.

Both portfolios show periods of weak performance as explained by negative Sharpe and Sortino ratios. This occurs especially during turbulent market conditions like 2021 and 2022. So neither strategy is able to consistently generate returns that compensate for the level of risk taken. This highlights an important limitation of momentum-based ensemble strategies. Whom is their sensitivity to changing market regimes and the presence of noise in financial data.

At the same time, the APW portfolio shows signs of improved efficiency in more favorable environments. For example, in 2025, the APW portfolio achieves a Sharpe ratio of 0.94 and the EW 0.11. Thus, the APW can enhance both return generation and downside risk management when market conditions are supportive.

These improvements are not uniform across the sample. The variance in both risk metrics and return distributions suggests that the performance of the adaptive model is highly sensitive to the signals. It depends heavily on their stability and consistency. In periods where signal performance is inconsistent or rapidly changing the adaptive model can amplify noise. This can reduce its effectiveness in improving decision-making.

Overall, the findings from this section indicate that while the APW portfolio can outperform the EW portfolio under certain conditions. The overall risk-adjusted performance remains limited. This suggests that the adaptive weighting approach improves relative performance but does not fully resolve the challenges associated with financial prediction and portfolio optimization.

5.3 Comparative Analysis of Portfolio Performance

This section provides a direct comparison between Equal-Weighted (EW) and Adaptive Performance-Weighted (APW) portfolios, integrating the results from the previous analyses to evaluate which strategy performs better, under what conditions, and to what extent.

From an overall return perspective, the results indicate that the APW portfolio achieves superior performance in several key periods, particularly in the later part of the sample. These differences could mean that the adaptive weighting is capable of enhancing performance when market conditions favor certain signals.

However, this outperformance is not consistent across all years. In some periods, the EW portfolio performs equally well or even better. For example, in 2020, the EW portfolio achieves a return of 8.44% while the APW -0.44%. So the adaptive model may underperform when recent signal performance is not indicative of future outcomes. The variation suggests that the effectiveness of the adaptive

strategy depends on the persistence of market trends.

The APW portfolio tends to perform better during periods with clearer and more stable trends. In these environments, the adaptive model is able to allocate more weight to signals that have recently demonstrated predictive strength. On the other hand, during periods of market stress or regime shifts, both portfolios experienced significant losses.

In terms of risk, both portfolios show similar overall characteristics with respect to drawdowns. For example, in 2021, both portfolios experience drawdowns with APW reaching approximately -10.03% and EW reaching -9.45%. Meaning the adaptive weighting does not alter exposure to market-wide downturns. However, in some periods, such as 2022, the APW portfolio demonstrates relatively smaller drawdowns compared to the EW portfolio. This suggests that the adaptive approach may provide partial protection in certain conditions.

A key distinction between the two strategies lies in the dispersion of returns. The APW portfolio has greater dispersion in both monthly and annual returns. So the adaptive model increases responsiveness to changing market conditions, allowing it to capture stronger positive outcomes. But this also exposes it to larger negative deviations. While the EW portfolio provides a more stable but less dynamic performance profile. This reflects the averaging effect of equal weighting across signals.

When considering risk-adjusted performance, the results suggest that neither strategy consistently dominates. While the APW portfolio achieves higher Sharpe and Sortino ratios in certain years, like in favorable market environments such as 2025, both portfolios exhibit negative risk-adjusted metrics in several periods. This indicates that although the adaptive model improves performance, it does not fully overcome the challenge. Those who are associated with achieving efficient risk-return trade-offs.

The findings suggest that the APW portfolio offers improved potential and performs better in trend-driven environments. But this advantage comes at the cost of increased volatility and does not consistently translate into superior performance across all periods.

In conclusion, the adaptive weighting approach enhances the ability to exploit favorable market conditions and improves performance in specific regimes. But this does not provide a universally superior solution. The Equal-Weighted portfolio remains a robust benchmark, offering more stable performance with lower sensitivity to short term fluctuation. These results support the view that dynamic programming signal weighting can improve portfolio performance. But its effectiveness is conditional on the market environment.

6 Discussion

This study set out to investigate whether weighting of trading signals would increase risk-adjusted returns. The results explain both the potential and the limitations of the adaptive weighting.

The findings indicate that the APW portfolio generates returns in several periods. These periods were the ones with clearer and persistent trends. This aligns with Chapter 3, where momentum-based strategies are expected to perform well when market signals show stability. By dynamically allocating higher weights to recently successful signals, the APW model is able to use these conditions more effectively than the EW benchmark.

The results demonstrate that this advantage is not consistent across all market regimes. During periods of instability or rapid shifts like 2021 and 2022, both portfolios experienced large losses in these periods. The adaptive approach does not provide enough protection and may even increase noise. Because it relies on recent performance that may not be indicative of future outcomes.

An important observation is that although the APW portfolio has higher returns in several years, the level of return over the full sample period remains somewhat small. This should not be mistaken as a failure of the model. The objective of this study is not to maximize absolute returns. It is to evaluate whether adaptive weighting provides an improvement over equal weighting. From this perspective, the results show that the adaptive approach offers improvements under certain conditions.

From the analysis, we observe the trade-off between responsiveness and stability. The APW portfolio shows greater variability in returns with both higher peaks and lower years. But the EW portfolio provides a more stable but less dynamic performance profile. This reflects the trade-off in portfolio construction. Increasing adaptability helps capture favorable opportunities. But it comes with the cost of increased fluctuations.

Finally, the results reveal the importance of parameter selection. The lookback window and prediction horizon change will vary the outcome a lot. The effectiveness of the adaptive model depends on the balance between these two.

A short lookback window may improve stability, but it can reduce adaptability. The prediction horizon matters because it determines how quickly the model reacts to changing conditions. These findings show that there is no single optimal setup.

6.1 Robustness/Sensitivity Analysis

To evaluate the stability of the APW model, we will conduct a robustness analysis. This will be done by varying the model parameters. The baseline configuration uses a threshold of 0.3, a prediction horizon of 10 days, and a lookback window of 120 days.

To assess the sensitivity of the model, we will test three alternative parameter configurations:

- Setup 1: Threshold 0.3, Horizon 5, Lookback 60
- Setup 2: Threshold 0.3, Horizon 10, Lookback 40
- Setup 3: Threshold 0.3, Horizon 20, Lookback 240

The purpose of this analysis is to determine whether the performance is robust across different parameters or if its dependent on the configuration.

6.1.1 Overall Performance Comparison

The results show that the relative performance of the APW and EW models varies. In some configurations the APW outperforms the benchmark, while in others the opposite happens.

In the short horizon configuration(Setup 1), both models deliver similar cumulative returns. Both provided limited returns in the highly reactive environments. In the medium horizon(Setup 2) with a shorter lookback window, the EW performs better. This could mean that the APW model overreacts to short-term noise when the evaluation window is small. Because in the long horizon configuration(Setup 3) the APW performs much better. This indicates that the adaptive weighting becomes more effective when signal performance is evaluated over a longer period.

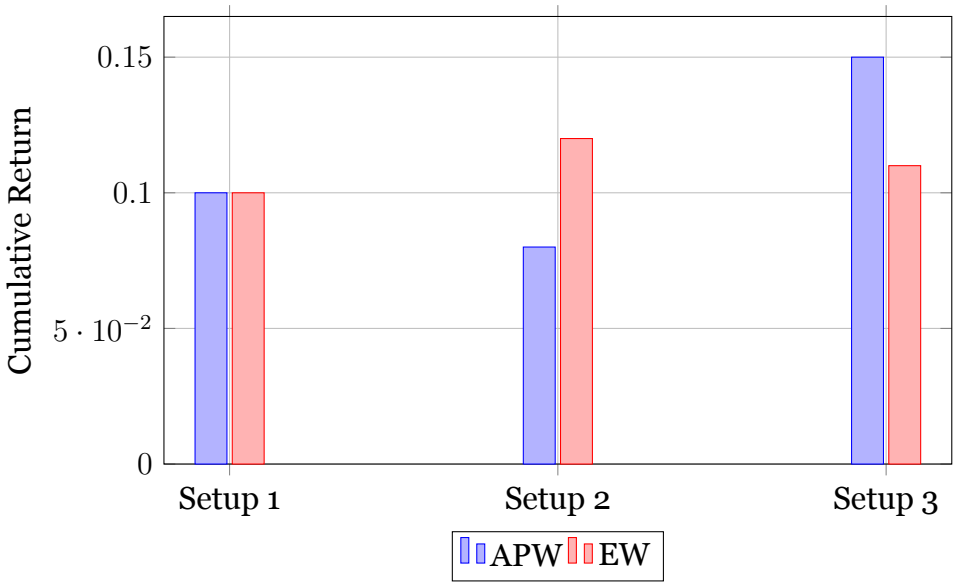


Figure 6.1: Cumulative return comparison between APW and Equal-Weighted (EW) models across different parameter setups.

6.1.2 Annual Performance Stability

A year-by-year analysis reveals that the APW model does not always outperform the EW. But we did notice that the outperformance increases in configurations with longer lookback windows.

As in Setup 3, the APW model outperforms the EW in a majority of years. While in Setup 2, the APW only performs better in a few periods. This could mean that the effectiveness of the APW model depends on the persistence of signal performance.

6.1.3 Risk-Adjusted Performance

The comparison of Sharpe ratios across parameters shows mixed results. In two of the three setups, the APW gives higher average Sharpe ratios. However, in the Setup with a shorter lookback window, the APW produces weaker Sharpe ratios. This could mean that the short evaluation periods introduce noise and reduce the robustness of performance based weighting.



Figure 6.2: Annual returns for APW and EW models across all years and parameter setups.

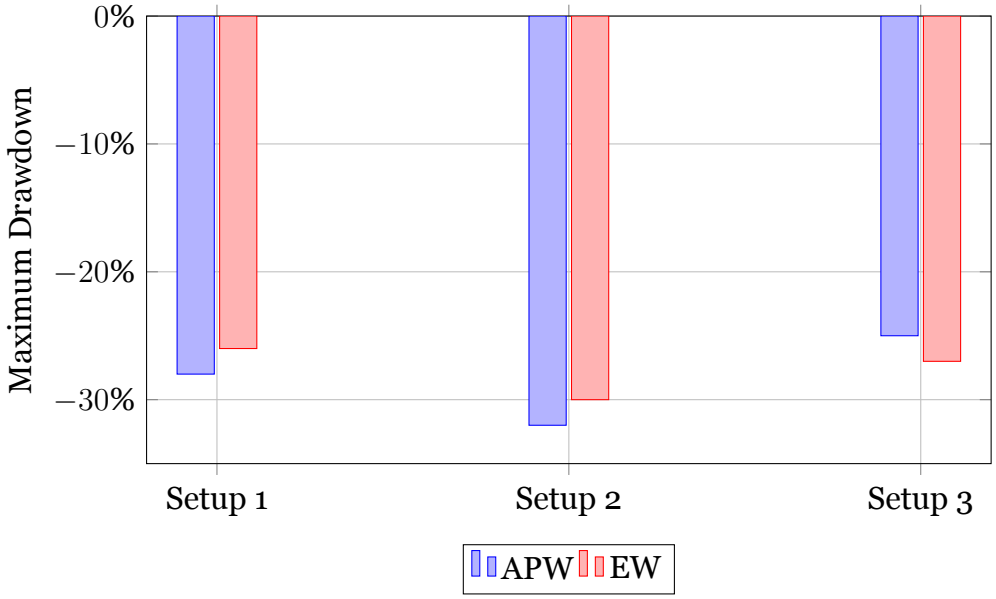


Figure 6.3: Maximum drawdown comparison between APW and EW models across parameter setups.

6.1.4 Drawdown and Downside Risk

The maximum drawdowns for the APW does not always reduce downside risk. In shorter configurations, the EW model shows slightly lower drawdowns. While in the longer configurations, the APW model provides marginal improvements.

This could mean that the APW does not change exposure to market downturns but instead offer improved stability when based on longer performance evaluation.

6.1.5 Parameter Sensitivity and Interpretation

From the above, we can notice that the model is sensitive to the choice of lookback window. Short lookback periods lead to unstable performance. This likely happens because of an overreaction to recent noise. On the other hand, longer lookback windows improve stability and allow the model to identify more persistent signal performance.

The prediction horizon also plays an important role. Longer horizons appear to reduce noise and improve the effectiveness of the APW model. And shorter horizons increase fluctuations in results. This could be because in longer periods,

the trend gets time to stabilize while in the shorter term the market can vary a lot.

Meaning shorter parameter settings allow model to react quickly to market changes, the also introduce performance fluctuations. Longer parameter settings improve consistency but reduce responsiveness.

7 Conclusions

7.1 Conclusion of Robustness Analysis

The robustness analysis shows that the adaptive weighting approach is not always superior to equal weighting. Its performance depends on the parameter selection and market conditions.

The results indicate that the APW can provide improvement when applied with a proper long evaluation window. And the proper window can be found by a trial-and-error approach. The performance-based signal weighting is a promising approach. But it requires careful parameter design to balance flexibility and stability.

7.2 Reliability

The reliability of the results is supported by several aspects of the study design. First, the analysis is based on a long observation period from 2017 to 2025. This includes multiple market regimes with periods of low volatility, high volatility, wars, epidemics and other shocks. This means that the results are not driven by a single specific market condition.

Second, using more assets and multiple momentum-based signals improves the validity of the results. The study combines signals across different asset classes and applies a consistent methodology. This approach reduces the risk that the findings are driven by idiosyncratic behavior in a single market instrument.

Third, the implementation of a rolling backtesting was in an out-of-sample setting. This was done to reduce the risk of look-ahead bias. However, some challenges

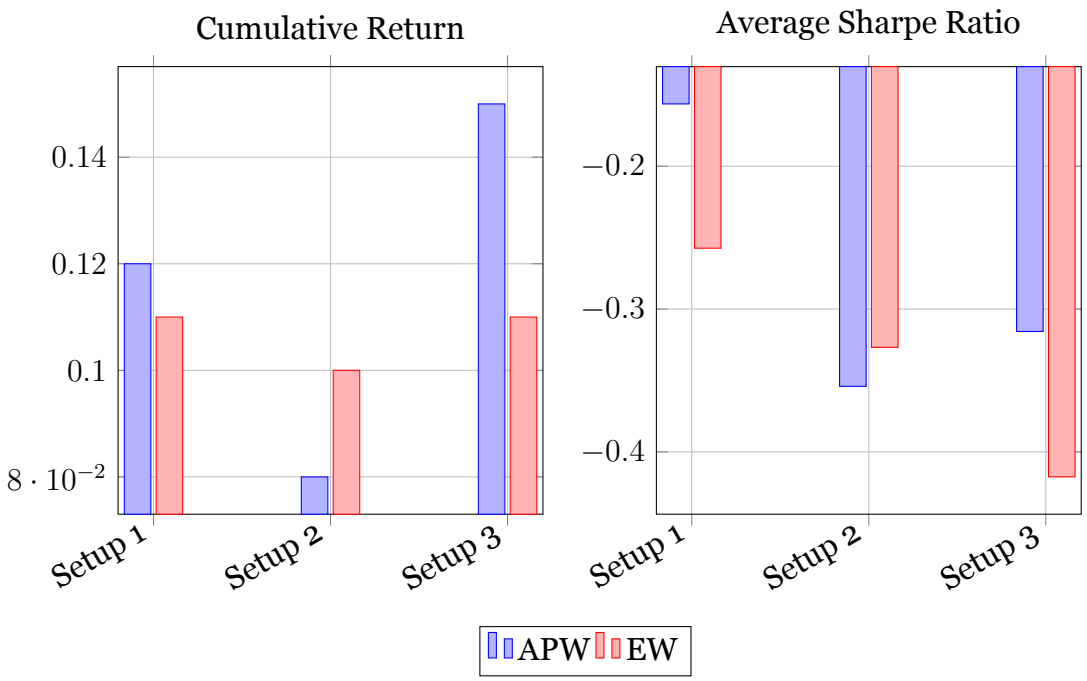


Figure 7.1: Summary comparison of cumulative return and risk-adjusted performance across parameter setups for APW and EW models.

were encountered during the data collection process that may affect reliability. For example, an initial dataset mistakenly included an incorrect instrument due to ticker name. A global bond ETF(AGGG) was confused with a different ticker (AGGG, Antilia Group, Cor.).

7.3 Limitations

Despite the strengths of the study, several limitations must be noted.

The analysis assumed no transaction costs, and liquidity constraints were not taken care of. These factors can have an effect on the portfolio performance, especially for strategies that involve frequent balancing. The lack of transaction costs may lead to an overestimation of real-world performance.

Second, the study uses historical data and assumes that past patterns are informative. But financial markets are non-stationary, and relationships observed in the past may not remain.

Third, the adaptive weighting based on relatively simple performance metrics and linear weighting schemes. While this enhances clarity, it may limit the

model's ability to capture more complex relationships between signals and market conditions.

Finally, the results are sensitive to the choice of parameters. Different choices may lead to different outcomes.

7.4 Future Work

Several directions for future research emerge from this study.

First, the adaptive weighting framework could be studied further by varying the key parameters in a systematic way. A more detailed robustness analysis would show how sensitive the model is to these choices. It could also help identify more stable parameter settings.

Second, the model could be extended by including transaction costs and other market frictions. This would give a more realistic view of performance. It would also help assess whether the strategy is practical to use in real markets.

Third, more advanced weighting schemes could be explored. For example, nonlinear weighting functions or machine learning approaches. These methods could help capture more complex relationships between signals and market conditions. But they may also increase the risk of overfitting, so a balance is needed.

Fourth, the framework could be tested on different assets or higher-frequency data. This would help evaluate whether the results can be replicated.

Finally, combining adaptive weighting with volatility targeting or dynamic position sizing may improve overall performance. It may also help reduce downside risk.

7.5 Final Words

This study examines whether an adaptive ensemble of trading signals can outperform a simple equal-weighted approach. The comparison focuses on returns and risk-adjusted performance.

The results suggest that the Adaptive Performance-Weighted portfolio can improve returns in some market conditions. This is true during periods of stable and persistent trends. But it is not consistent over time. Both portfolios show variation in performance and remain vulnerable to adverse market conditions. Risk-adjusted performance remains limited, as reflected in negative Sharpe and Sortino ratios in several years.

The findings suggest that adaptive weighting provides a relative improvement compared to equal weighting. The effectiveness depends on the stability of the underlying signals and the market conditions. In conclusion, adaptive ensemble methods offer a promising approach to improving portfolio performance. However, their benefits are conditional and must be evaluated in the context of both market dynamics and model design.

References

Markowitz, H. (1952). Portfolio Selection. *The Journal of Finance*, 7(1), pp. 77–91.

DeMiguel, V. Nogales, F.J. (2009). Portfolio Selection with Robust Estimation. *Operations Research*, 57(3), pp. 1–3 (PDF).

Bodnar, T., Niklasson, V. Thorsén, E. (2021). Volatility Sensitive Bayesian Estimation of Portfolio VaR and CVaR. *Mathematical Statistics Research Report*, Stockholm University, pp. 1 (PDF).

Rodriguez Dominguez, A., Shahzad, M. Hong, X. (2025). Multi-hypothesis prediction for portfolio optimization: A structured ensemble learning approach. *Expert Systems with Applications*, pp. 1–2 (PDF).

Brooks, J. (2017). A Half Century of Macro Momentum, pp. 3–5 (PDF).

Hedlund, J. (2024). Portfolio Optimization using Deep Reinforcement Learning Models. Master's thesis, Lund University, pp. 2, 6 (PDF).

Lassance, N., Martín-Utrera, A. Simaan, M. (2023). The Risk of Expected Utility under Parameter Uncertainty, pp. 3–5 (PDF).

Schmid, O. Wirth, P. (2021). Optimal Allocation to Time-Series and Cross-Sectional Momentum. *The Journal of Portfolio Management*, 47(4), pp. 3–5 (PDF).

Appendices

Appendix A: Asset Universe

Asset Class	Ticker	Description / Underlying Index	Region
Equities	SPY, IVV, VOO	S&P 500	USA (Large Cap)
	QQQ	Nasdaq 100	USA (Technology)
	IWM, VTWO	Russell 2000	USA (Small Cap)
	FEX	Euro Stoxx 50	Europe (Large Cap)
	VGK	Euro Stoxx 600 / MSCI Europe	Europe (Broad)
	EWD	MSCI Sweden (OMX Proxy)	Sweden
	EWG, EWQ	MSCI Germany, MSCI France	Germany, France
	EWI, EWU	MSCI Italy, FTSE 100 (UK)	Italy, United Kingdom
	EWJ, SCJ	MSCI Japan, MSCI Japan Small Cap	Japan
	VTI, ITOT	Total Stock Market	USA (Total)
	URTH, ACWI	MSCI World, MSCI ACWI	Global
	EEM, VWO	MSCI Emerging Markets	Emerging Markets
	MCHI, FXI	MSCI China	China
	INDA, EWZ	MSCI India, MSCI Brazil	India, Brazil
	EWL, EWC, EWA	MSCI Switzerland, Canada, Australia	CH, CA, AU
Sectors	VNQ	Global REIT	Real Estate
	IBB, XBI	Nasdaq Biotechnology	Healthcare/Biotech
	ICLN, INRG	Global Clean Energy	Renewable Energy
Currencies	UUP, FXE	US Dollar, Euro	Currency
	FXY, FXB	Japanese Yen, British Pound	Currency
	FXF, FXC	Swiss Franc, Canadian Dollar	Currency
	FXA, CYB	Australian Dollar, Chinese Yuan	Currency
Fixed Income	BND, AGG	US Total Bond Market	Broad Bond Market
	SHY, IEF, TLT	US Treasuries (1-3Y, 7-10Y, 20+Y)	Government Bonds
	TIP, SCHP	US TIPS	Inflation-Protected
	LQD, HYG, JNK	Corporate & High Yield Bonds	Corporate Credit
	BNDW	Global Aggregate Bonds	Global Bonds
	EMB	Emerging Market Bonds	EM Debt
	BSV, IGIB, FLOT	Short-term & Floating Rate Bonds	Specialized Fixed Income
Commodities	GLD, IAU	Gold	Precious Metals
	SLV	Silver	Precious Metals
	USO, UNG	Crude Oil, Natural Gas	Energy
	CPER, URA	Copper, Uranium	Industrial Metals/Energy
	WEAT	Wheat	Agriculture

Table .1: Asset universe consisting of 65 instruments across asset classes.

Kirollosbo (2026). Bachelors-thesis-excel-code-and-graphs-. GitHub repository. Available at: https://github.com/kirollosbo/Bachelors-thesis-excel-code-and-graphs- (Accessed: 2 May 2026)

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